Huolitsija operating and go-live runbook

25 October 2026 · Road Step 7 · live identity `f53f2fa06ba444c47e68e7efc05c8087c8dd021f` (since 01:44 UTC). All five legal pages, including the DPA, are verified on this identity; the recorded journey is still gated.

Start with Owner activation actions: exact company steps, integrator handoff and required receipts for all ten provider areas. It supersedes older activation guidance below where noted.

Use Autopilot for routine operations, with an accountable owner and qualified contracted specialists for the legal responsibilities listed in LEGAL-HUMAN-POINTS.md. This document does not certify commercial readiness. Several integrations remain partial or unconfigured; production Customs requires transport and customer-conformance evidence, not merely credentials.

Latest acceptance rule: capability rows use only **live** / **live-ready** / **partial** / **missing**. Test mode is permitted only when production is blocked on a named company-only step (incorporation, signed contract or payment). Missing implementation or unproven behavior stays partial/missing; simulated success is not a final capability result.

Evidence and revision discipline

Before any live run, read `https://huolitsija-app.emergent.host/api/version`, record its commit and the UTC time, and compare the rendered frontend asset identity. Read authenticated `/api/integrations/status`, `/api/settings`, `/api/autopilot/status`, the sourcing status and insurance provider status from the same release. Use the credential-store **sign_in** workflow for demo accounts; never print passwords, session cookies, API keys, webhook tokens or PEM keys. **Correction:** an earlier claim that demo passwords were already in the person's store was wrong. Presence and a successful sign-in must be observed. The corrected review path is `evidence/s2-review/r2/REVIEW-r2.md`.

Integrator's live probe at 21:53 UTC against `a86ad24`: email `log`, SMS `log` and disabled; OpenAI available; Stripe test mode and webhook enabled; Tulli disabled; insurance provider absent (Redkik sandbox unconfigured); credit bureau absent; CRM and partner messages use sandbox domains and log delivery; company Business ID, VAT ID and EORI absent. These are a dated baseline, not current proof after another deploy. Hosting is Emergent, with MongoDB through Emergent; region and contractual subprocessors are unverified.

Current live state — f53f2fa, 25 October 2026 02:05 UTC

Read from the public host at 02:05 UTC; it supersedes the r8a notes below where they differ.

Area	Observed on `f53f2fa`	Evidence
Legal pages	`/terms` (version 1.1), `/privacy`, `/dpa`, `/cookies` and `/company` render in Finnish and English, light and dark, at 1440 and 390 px: 40 of 40 checks pass, no horizontal overflow, the DPA shows its seven sections. The r8a finding that the DPA was absent no longer applies.	`evidence/s7-legal-live-f53f2fa/results.json` and its 48 screenshots

Area	Observed on `f53f2fa`	Evidence
Company identity	`GET /api/legal/config`: name Huolitsija Oy, `registered=false`; Business ID, VAT ID, EORI, address, phone and registered office are empty; contact and privacy e-mail `ops@huolitsija.fi`; notice version 1.1.	live probe 02:05 UTC
Autopilot	`GET /api/autopilot/status`: enabled=false, halted=false (last changed 00:45 UTC, incident drill). It stays off globally; only a test-tenant switch is allowed.	live probe 02:05 UTC
Runbook in the app	`/ops/runbook` (internal staff only) serves this runbook, its companion documents and this PDF from the release. Handed to the integrator as `18fc522` for the 02:30 UTC train; it counts as live only once `GET /api/runbook` answers on the public host.	branch `feat/runbook-0230`
Recorded journey	Not run. Gates: s2's email hop on test tenant `e19d517b`, the master's EUR.1 clearance and the integrator's activation.	—
Capability count	14 of 104 rows live (recount lane, sprint state 02:00 UTC).	`shared/capmap/CAPABILITY-MAP-live.md`

r8a observed state — 25 October 2026

Integrator evidence at 00:27 UTC identifies `fc211b4ec3514c733c719367881563c2f5858004`; an independent s4 version read on this continuation matches. Source: `shared/evidence/integrator-r8/r8a-live/api-probes.json`. Deep health returned HTTP 200 with all listed checks `ok`; this is service health, not customer-journey proof. Email is Postmark live with sender `Huolitsija <marc+huolitsija@heliosone.fi>`; Stripe remains test mode with its webhook enabled. SMS is disabled/log-only. Tulli is disabled with no endpoint or declarant ID configured. The company remains unregistered with Business ID, VAT ID, EORI, address and registered office missing. Contact and privacy email are `ops@huolitsija.fi`.

The integrator's 00:28:48 UTC Autopilot read shows enabled=false and halted=false. Keep it off until the master confirms the s8 EUR.1 test is done and permits activation; the integrator must also confirm the single-RFQ guard and webhook/config readiness. Only the integrator enables it immediately before recording. Prospecting/sourcing outreach stays log-only. Step 7 observes and records; it does not toggle Autopilot or manually advance business work.

The shared capability recount for this exact SHA is **12/104 live, 2 live-ready, 85 partial, 5 missing**. This is the capability-map lane's count, not a new s4 recount. Step 7 has no standalone rows in the fixed 104-row inventory; publishing legal pages does not increase the numerator. No road step is complete solely because its UI renders.

The earlier four-route legal proof is bound to r7 `00c10d5`; do not relabel its screenshots as r8. The s4 eight-state `/dpa` probe against this identity rendered the homepage in every state, using frontend asset `main.a7b80349.js`; the expected DPA heading and seven sections were absent. DPA source exists in `66ada89`, but its deployed

publication is unverified pending the integrator's final r8 SHA. Failure evidence: `evidence/s4-continuation/live-r8-dpa/results.json`. Company fields stay incomplete regardless of legal-page publication. The recorded digital journey and separate Stripe test-paid receipt remain unverified until their actual evidence is retained.

Launch sequence

1. Incorporate and record the legal entity and authorised representatives. Populate identity below; obtain tax registration, EORI and relevant mandates. Do not borrow Helios One Oy's Business ID for the new company.
2. Name the responsible privacy/controller contact and statutory decision owners. Confirm actual processor contracting entities, DPAs, hosting/data regions, international-transfer safeguards and retention policy. Test the contact channel. An unincorporated name is not a complete GDPR controller identity.
3. Establish approved mail delivery and inbound routing, OpenAI configuration, authorised payment, carrier and customs paths, monitoring and backups. Preserve isolation between test tenants and commercial customers.
4. Run FINAL-RUN-PLAN.md on the exact deployed release. Gate on business behavior, not passing unit tests. Capture all manual actor interventions.
5. Activate contracted live accounts individually, verify their receipts, then enable Autopilot and sourcing within approved limits. For an unavailable integration keep its explicit unavailable state; never convert a stub response to success.
6. Commercial launch still requires actual company identity, service contracts, insurance arrangements and any required Customs/ICS2 certification. Record these as evidence, not placeholder checkmarks.

Configuration ownership

Set secret values through the deployment's existing secret store. Backend env changes require a backend restart/redeploy; `REACT_APP_BACKEND_URL` is a frontend build-time value. Settings are persisted in MongoDB (`key=app`); a new environment default does **not** replace an existing stored setting. `/api/settings` requires an internal session; writes require `settings.write`. Never export that whole response into public evidence.

CONFIG-INVENTORY.md enumerates every literal runtime environment name and every core app setting in this source snapshot, with owning code and defaults where declared. Dynamic families and pending-step variables are expanded below. PROCESSORS.md maps recipients to real code and observed activation. Re-run the inventory after other branches land; this is part 1, not a claim that unpublished branch configuration is deployed.

Published terms are now version 1.1; new signup consent records use 1.1. Preserve existing contract versions. `TERMS-v1.0-source.js` retains the prior published wording for historical records; it is not the current notice. If stored `terms_version` remains 1.0, update the setting to `1.1 · NSAB 2015 (PSYM 2015)` for new quotes after deployment, without rewriting existing quotes/invoices.

Platform, identity and authentication

Variables / switches	Purpose and go-live action
`MONGO_URL`, `DB_NAME`	Persistent database connection/database; secret URI. Use a separate disposable DB for tests. Obtain backup/restore policy from Emergent and prove a restore in isolation.

Variables / switches	Purpose and go-live action
`APP_BASE_URL`, `REACT_APP_BACKEND_URL`, `CORS_ORIGINS`	Public links/payment redirects, optional split API URL, exact allowed origins. Production must use HTTPS and restrict origins; avoid `*`. Same-origin `/api` is the normal frontend path.
`JWT_SECRET`, `SESSION_DAYS`, `COOKIE_SECURE`	Stable protected signing key (or existing DB-stored key), default 14-day maximum session, secure cookie true in production. Key rotation can invalidate sessions; record and test it.
`COMPANY_NAME`, `COMPANY_BUSINESS_ID`, `COMPANY_VAT_ID`, `COMPANY_EORI`, `COMPANY_ADDRESS`, `COMPANY_EMAIL`, `COMPANY_PHONE`, `COMPANY_IBAN`, `COMPANY_BIC`	Registered name, Y-tunnus, tax/customs IDs, full postal address, monitored mailbox, optional telephone and settlement account. Set from authoritative incorporation/bank records. `seller_business_id`, `seller_vat_id`, `seller_iban`, `seller_bic` stored settings override corresponding defaults. Reconcile documents and public identity.
`COMPANY_REGISTERED`, `COMPANY_REGISTERED_OFFICE`, `COMPANY_PRIVACY_EMAIL`	s4 additions: explicitly confirm registration, registered municipality and monitored privacy contact (falls back to company email). `/api/legal/config` exposes only public identity, session lifetime and notice version. An ID alone does not imply incorporation.
`SEED_DEMO`, `DEMO_OPS_EMAIL`, `DEMO_OPS_PASSWORD`, `DEMO_RESET_PASSWORDS`	Seed/test identity controls. Disable demo seeding/reset for commercial operation; preserve needed test identities in their isolated sandbox. Do not reset existing customer passwords.
`API_RATE_LIMIT_PER_MIN`, `PUBLIC_QUOTE_RATE_LIMIT_PER_HOUR`, `PRIVACY_RATE_LIMIT_PER_EMAIL_HOUR`, `PRIVACY_RATE_LIMIT_PER_IP_HOUR`, `TRUSTED_PROXY_HOPS`	Abuse controls. Configure trusted proxy hops to the real ingress, so spoofed forwarding headers cannot bypass client limits.

Autopilot, emergency stop and daily operations

Enable at Ops → Autopilot (or the Settings autonomy control), using the exact confirmation `enable`. The endpoint is `POST /api/settings/autonomy/autopilot_enabled` with `{ "enabled": true, "confirmation": "enable" }`. This is a privileged, audited action; verify it first on a controlled authorised company journey. Set guardrails before enabling.

Runtime / setting	Operational meaning
`JOBS_WORKER_ENABLED`, `JOBS_POLL_SECONDS`, `JOBS_CONCURRENCY`	Durable worker on/off; default polling 2 seconds, concurrency 4. A running process does not prove jobs advance; inspect jobs, leases and heartbeat.
`autopilot_enabled`, `autopilot_halted`	Enabled is business automation; halted is global hold. Do not set these by raw DB writes. Agent jobs stay queued/retry while halted.
`guard_spend_per_day_cents`, `guard_confidence_floor`, `guard_claim_limit_cents`, `guard_default_credit_limit_cents`	Daily committed spend, confidence, claims and customer exposure. Currency is cents; verify boundary behavior in the actual decision log.
`insurance_claim_limit_cents`	Insurer-delegated settlement ceiling. Default zero means no authority recorded. Do not raise without the policy/insurer mandate.
`owner_alert_email`, `OWNER_ALERT_EMAIL`, `owner_queue_sla_hours`, `owner_escalation_hours`	Owner notification/digest and escalation. Confirm actual delivery; log mode is not an alert to a person.

Runtime / setting	Operational meaning
`customs_filing_delegated`, `customs_delegation_scope`, `customs_confirm_timeout_hours`	Internal standing filing delegation and scope (`direct` or `direct_and_indirect`); customer POA remains separately required. Use `POST /api/autopilot/customs-delegation` with enabled/scope/confirmation/signer_name/signer_title.
`TEST_TENANT_SANDBOX`, `SIMULATOR_ENABLED`, `SIMULATOR_STEP_SECONDS`	Reserved test-company path and simulated movement. Never use simulation as proof of physical freight movement. Simulated test-tenant results never prove a real provider transaction.
`EXC_REBOOK_HOURS`, `DELIVERY_POD_WAIT_HOURS`, `INVOICE_CUSTOMS_WAIT_HOURS`, `INVOICE_AUTO_ISSUE`, `EXCEPTION_DETECTION`	Exception and delivery/invoice timing controls; POD and customs evidence must still be present.

Emergency stop: use Ops → Autopilot stop / `POST /api/autopilot/kill`; inspect `autopilot_halted=true` and the held job list. Stop sourcing separately with `POST /api/partner-network/stop` (cancels queued sends) and, if necessary, `PARTNER_NETWORK_KILL=true`. Stop CRM outbound execution as well via its documented enabled/Autopilot control and provider log mode. This does not recall already sent email, cancel a bank payment, undo a carrier booking or retract an accepted Customs declaration. Inspect in-flight jobs and external receipts before retrying; uncertain external submissions require reconciliation, never blind replay.

Resume: reconcile in-flight effects, correct the incident, then `POST /api/autopilot/resume` with confirmation `enable`. Read the returned switches and explicitly enable each desired autonomy switch; do not assume sourcing resumes. Run one controlled transaction before widening traffic.

Daily owner check: failed/late jobs and dead letters; approvals and legal holds; missing carrier confirmations, customs responses and POD; overdue invoices/payment mismatches; message bounces; sanctions/PEP source freshness; retention failures; spend and credit guards. Retry a dead letter only after determining whether the provider already acted. Review the daily digest as monitoring evidence, not as proof that all business work succeeded. A sole owner can contract cover for required human decisions during absence.

OpenAI provider and model

`LLM_PROVIDER=openai`, `OPENAI_API_KEY`, `OPENAI_BASE_URL` (official API base), `LLM_MODEL`, `LLM_MODEL_FAST`, `LLM_REASONING_EFFORT`, `LLM_MAX_CONCURRENCY`; legacy `LLM_VISION_PROVIDER`/`LLM_VISION_MODEL`, `EMERGENT_LLM_KEY` and `ANTHROPIC_API_KEY` exist in the base snapshot. **Use OpenAI only.** Do not activate legacy alternate-provider fallbacks. The model-fix lane owns routing changes; inspect its final code and runtime status before launch.

Fetch the live account's `/v1/models` using the protected OpenAI key; save only model IDs and timestamp, never headers/key. Select an ID from that response, then prove text, structured JSON and document/image behavior through the actual app. A models-list result alone does not prove the chosen model supports the requested payload. `scripts/llm_check.py --list` is the repository's diagnostic; review its current implementation before execution. No model ID is prescribed from memory in this runbook. Verify `ai_runs` records OpenAI for each relevant operation; check the final vision branch explicitly.

Email, SMS and inbound journeys

Variables / switches	Activation and observable proof
`EMAIL_PROVIDER`, `EMAIL_FROM`	`log` stores only; `postmark`, `resend`, `smtp` use their actual delivery adapters. Choose a verified sender at heliosone.fi until a company domain is available.
`POSTMARK_SERVER_TOKEN`, `POSTMARK_WEBHOOK_TOKEN`	Use the existing server, request/confirm account approval, verify **DKIM and Return-Path on heliosone.fi** from Postmark's generated DNS values (do not guess selectors). Test-domain delivery is not approval to send to arbitrary recipients. Configure Delivery/Bounce webhooks on `/api/webhooks/postmark`; token is HTTP Basic-auth secret. Confirm delivered event and external inbox receipt.
`RESEND_API_KEY`, `RESEND_WEBHOOK_SECRET`	Alternative adapter, only if actually selected/contracted. Signature-verified delivery events; avoid claiming it is currently in use because code exists.
`SMTP_HOST`, `SMTP_PORT`, `SMTP_USER`, `SMTP_PASSWORD`, `SMTP_SECURITY`	General SMTP (`starttls` or `ssl`; `none` only isolated tests). Separate `PARTNER_SMTP_*`, `CRM_SMTP_*` share these five suffixes.
`SMS_PROVIDER`, `TWILIO_ACCOUNT_SID`, `TWILIO_AUTH_TOKEN`, `TWILIO_FROM` or `TWILIO_MESSAGING_SERVICE_SID`, `TWILIO_STATUS_CALLBACK_URL`	`log` sends nothing. **Trial → paid:** upgrade the existing Twilio account, assign an SMS-capable sender/service, complete destination-country registration, use live account credentials. Trial verified-recipient restrictions and test-credential magic numbers are not production. Callback `/api/webhooks/twilio/sms-status`; prove real delivery to an opted-in test recipient and the authenticated status callback.
`sms_notifications_enabled` and customer opt-in	Both operational switch and recipient preference must permit service SMS. Turning the global switch on does not grant customer consent.
`EMAIL_CHANNEL_IMAP_HOST/PORT/USER/PASSWORD/SECURITY/FOLDER`, `EMAIL_CHANNEL_WEBHOOK_SECRET`, `EMAIL_CHANNEL_FROM`, `EMAIL_CHANNEL_DOMAIN`, `EMAIL_CHANNEL_PROVIDER`	s6 declared inbound/outbound email-only path. Raw RFC822/Postmark input `/api/email-channel/inbound` uses HMAC of raw bytes. Verify actual provider support or signed relay; never disable authentication to make Postmark work. Message-ID correlation and sender matching must survive retries.
`email_channel_enabled`, `email_channel_auto_reply`	s6 proposed settings; reconcile after its merge. An email-only journey may use secure one-time KYC links and Stripe-hosted checkout, but must not require portal login or an ops operator to advance routine work.

Partner sourcing and CRM

Sourcing is a separate switch: `POST /api/settings/autonomy/partner_network_enabled` with `{ "enabled": true, "confirmation": "enable" }`; Ops → Sourcing / Hankinta. It is independent of routine Autopilot, while respecting global halt. Limits: `partner_target_margin_pct`, `partner_max_price_cents`, `partner_sell_price_cents`, `partner_approved_lanes`, `partner_daily_send_limit`, `partner_max_rounds`, `partner_rfqs_per_lane`, `partner_counter_anchor_pct`. Keep an approved-lane list and demonstrate that opt-outs prevent later sends.

PARTNER_EMAIL_PROVIDER, PARTNER_EMAIL_FROM, PARTNER_REPLY_TO; PARTNER_IMAP_HOST/PORT/USER/PASSWORD/SECURITY/FOLDER; PARTNER_INBOUND_SECRET, PARTNER_INBOUND_REQUIRE_AUTH, PARTNER_OPTOUT_SECRET; PARTNER_SANDBOX_DOMAIN, PARTNER_NETWORK_KILL; optional PARTNER_SMTP_*. **Remove** PARTNER_SANDBOX_DOMAIN **only after approved sender/inbound delivery, opt-out handling, limits and one controlled real outreach are proved.** Empty/unset changes the recipient from a sandbox mailbox to the actual carrier. Stop queued messages and inspect recipients before switching.

CRM: CRM_EMAIL_PROVIDER, CRM_EMAIL_FROM, CRM_REPLY_TO, CRM_PUBLIC_BASE_URL, CRM_OPTOUT_SECRET, CRM_PROSPECT_DAILY_AT, CRM_IMAP_HOST/PORT/USER/PASSWORD/SECURITY/FOLDER, CRM_SMTP_*, CRM_SANDBOX_DOMAIN, TRADE_STATS_SOURCE. Settings: crm_daily_send_limit, crm_follow_up_days, crm_max_follow_ups, crm_target_industries, crm_target_regions. **Remove** CRM_SANDBOX_DOMAIN **separately** after confirming sender, lawful business contact selection, source attribution and opt-out. Removing one sandbox domain does not remove the other.

Quotes, credit and insurance

QUOTE_APPROVAL_THRESHOLD_EUR, DEFAULT_PAYMENT_TERMS_DAYS, QUOTE_CUSTOMS_INCLUDED_ITEMS, QUOTE_ICS2_FEE_EUR, QUOTE_CUSTOMS_FEE_VAT_CODE, QUOTE_DOMESTIC_LEG_VAT_CODE, PUBLIC_QUOTE_RATE_LIMIT_PER_HOUR. Core settings include margins, clearance/document fees, quote validity, VAT rates, payment terms, credit ceilings and autopilot_quote_max_cents/autopilot_min_margin_pct; see inventory. Revalidate legal tax rates at launch rather than treating source defaults as permanent tax advice.

Global geocoding/routing: GEOCODER_URL, PHOTON_URL, OSRM_URL, SEA_AVOID_RED_SEA. Distinguish geocoder coordinates, rate-card evidence, live carrier rates, indicative estimates and unanswered RFQs. An absent seeded rate must not be replaced with an invented confirmed price.

Credit: CREDIT_BUREAU=none|asiakastieto|creditsafe; ASIAKASTIETO_BASE_URL/USER_ID/PASSWORD, CREDITSAFE_BASE_URL/USERNAME/PASSWORD; settings credit_default_limit_cents, credit_auto_limit_max_cents, credit_hold_overdue_days. A PRH/YTJ lookup is not a bureau credit report. Personal/sole-trader decisions need the privacy assessment in the legal matrix.

Insurance: INSURANCE_PROVIDER=redkik takes precedence over setting insurance_provider; REDKIK_BASE_URL, REDKIK_CLIENT_ID, REDKIK_CLIENT_SECRET, REDKIK_POLICY_ID, REDKIK_COMMODITY_ID, REDKIK_COVER_TYPES. Obtain a real distribution/policy arrangement and organisation credentials. Default base is https://sandbox.app.redkik.com; production is https://sales.app.redkik.com in the adapter. Verify these against the granted account before switching. Test quote → explicit bind → real policy ID/certificate and limits. stub_named is **not insurance**. Loadsure is documented-only in this snapshot, despite an older interface draft suggesting a provider switch. Settings insurance_rate_pct/insurance_min_premium_eur do not create coverage.

Carrier, execution, documents and visibility

CARRIER_BOOKING_MODE=log|sandbox|live, CARRIER_CONFIRM_HOURS; dynamic CARRIER_<CODE>_BASE_URL, _BOOKING_URL, _API_KEY, _ENABLED, _SANDBOX, _PORTAL_ACCOUNT, _SERVICE_CONTRACT_REF, _CONTRACT_REF. Codes are normalised from carrier records; adapter type selects required names. Test a real contract/quotation reference and a correlated confirmation; a request acceptance is not a confirmed booking. DCSA_<SCAC>_BASE_URL/API_KEY/API_SECRET/KEY_HEADER is the separate tracking family. DHL_API_KEY, DIGITRAFFIC_ENABLED, POSITION_TICK_SECONDS control existing tracking inputs.

s9 declared: EXECUTION_FROM, EXECUTION_SANDBOX_DOMAIN, WMS_PROVIDER=none|email|odoo, WMS_ODOO_URL/DB/USER/API_KEY; operations include SI/VGM cutoffs, pickups, warehouse handoff and consolidation. Check the actual merged reader before setting. Sandbox recipient rewrites must be removed only after the same controlled-delivery checks as sourcing.

s8 declared: TRADEDOCS_AUTO, EBL_PLATFORM=none|dcsa, EBL_ISSUANCE_URL, EBL_ISSUANCE_TOKEN, COO_CHAMBER_EMAIL, TRADEDOCS_ISSUER_*. A generated B/L PDF is not a carrier-signed negotiable title; an origin application is not a chamber-issued certificate. Record actual issuers/signatures and evidence.

s10 declared: AISSTREAM_API_KEY, AIS_GLOBAL_ENABLED, FLIGHT_TRACKING_PROVIDERS, OPENSKY_CLIENT_ID/CLIENT_SECRET, TERMINAL49_API_KEY, MILESTONES_FROM_SCHEDULE, EXC_ROLLOVER_DAYS, EXC_CONGESTION_HOURS. Record provenance/freshness. A map route, AIS vessel position or scheduled milestone is not proof that a specific shipment was physically delivered.

Customs, trade compliance and retention

TULLI_MODE=disabled|sandbox|test|production; TULLI_IMPORT_MODE, TULLI_EXPORT_MODE, TULLI_TRANSIT_MODE override per system once s11 lands. TULLI_ENDPOINT, TULLI_CLIENT_CERT, TULLI_CLIENT_KEY, TULLI_DECLARANT_ID; sandbox timing TULLI_SANDBOX_RELEASE_SECONDS, TULLI_SANDBOX_DECISION_SECONDS, and s11 TULLI_SANDBOX_EXIT_SECONDS.

Production transport remains partial pending deployed authority proof. Customs lane reconciliation of 9258593..9d7e95f reports no transport/configuration changes: the v1.21 implementation already in 9258593 passed 40 protocol and 5 exchange-durability checks in the frozen full run. The supplied schemas/examples and guidebooks are available; no acquisition request remains. No deployed company transport or live authority acceptance was verified. UTU customer status through Authorisations and Decisions, separate DVV enrolment/certificate, applicable procedure testing and customer mandates remain distinct gates. Local protocol/XSD checks do not prove Customs acceptance.

Transit additional names: TULLI_HOLDER_EORI, TULLI_HOLDER_NAME, TULLI_TRANSIT_OFFICE_OF_DEPARTURE, TULLI_GUARANTEE_GRN, TULLI_GUARANTEE_ACCESS_CODE. Guarantee amount/currency is declaration data. EU COL office ingestion is incomplete; fresh verified office records/roles are required. Arrival and unloading require actual physical-event evidence.

ICS2: ICS2_MODE=disabled|sandbox|sti|conformance|production; ICS2_EORI, ICS2_AS4_ENDPOINT, ICS2_AS4_CERT, ICS2_AS4_KEY, ICS2_PARTY_ID, ICS2_RECEIVER_PARTY_ID, ICS2_SERVICE, ICS2_ACTION, ICS2_FINAL_RECIPIENT, optional ICS2_PARTY_ID_TYPE, ICS2_SERVICE_TYPE, ICS2_AP_USERNAME, ICS2_AP_PASSWORD. Obtain EU Login/UUM&DS roles, registered AP/PMode and self-conformance approval. AP acknowledgement differs from ENS business acceptance. A downloaded STI XML file is not a submitted declaration. The implemented Domibus profile is inline/base64; MTOM retrieval is unimplemented. See shared/s11/golive-ncts-ics2.md.

TARIC_SOURCE=consultation is the documented source; HTTPS access plus DB is required, no paid key. Do not present an AI fallback estimate as authoritative duty. customs_four_eyes_duty_eur, customs_four_eyes_value_eur, customs_authorised_filers, customs_rescreen_days are s11 settings. Keep authority, POA, EORI and independent-review predicates intact.

Compliance: TRADE_GOV_API_KEY optional; OPENSANCTIONS_API_KEY needs licence; COMPANIES_HOUSE_API_KEY; actual current PRH reader uses PRH_BO_API_URL, PRH_BO_CLIENT_ID, PRH_BO_USER_ID, PRH_BO_SERVICE (the older interfaces.md mentions a client secret, so reconcile the contract before provision). IATA_DGR_DATA_PATH is a licensed data path, not permission to redistribute a manual.

Retention settings: `record_retention_years`, `retention_financial_statements_years`, `retention_quote_unconverted_months`, `retention_kyc_years_after_relationship`, `retention_token_days`, `retention_privacy_request_years`. Review `/api/compliance/retention/*`, legal holds and export/erasure requests. Run dry-run first, inspect real candidate records and law/purpose, then apply. Global halt may hold scheduled retention; owner must cover deadlines. Customer access closure does not erase mandatory accounting/customs records. Backups need their own retention/restore controls.

Billing, payments, finance, API and EDI

`STRIPE_API_KEY` determines test/live; `STRIPE_WEBHOOK_SECRET` or Emergent-compatible `PAYMENT_WEBHOOK_SIGNING_SECRET` must match the same Stripe environment and endpoint `/api/webhooks/stripe`. Use a real authorised payment; test mode is permitted only while the named company activation step blocks live payments. Go-live: complete Stripe business verification with the real representative, use live keys and live webhook secret, check amount/currency/idempotency and reconcile a controlled payment. A checkout return URL is not payment proof; authenticated webhook and ledger allocation are required.

`TRACKING_WEBHOOK_SECRET`, `BANK_WEBHOOK_SECRET` sign raw provider payloads. `/api/webhooks/bank/camt054` imports XML; verify exact bank line, invoice matching, duplicates and unmatched handling. Dunning settings: enabled, first/second/final/collection days, fixed compensation and reminder fee. Do not treat the source's fee/interest citation as legal approval; validate current interest/fee rules with the actual B2B contract.

s13 declared: `FX_SOURCE`, `AP_IMAP_HOST/PORT/USER/PASSWORD/SECURITY/FOLDER`, `AP_INBOUND_REQUIRE_AUTH`, `BANK_PAYMENT_MODE=file|api`, `BANK_API_PROVIDER` and provider-specific `BANK_API_*`, `LEDGER_EXPORT_FORMAT=procountor|netvisor|csv`, `LEDGER_API=off|netvisor|procountor` and account keys, `COLLECTIONS_AGENCY=none|file`, `VAT_FILING_MODE=file|api`. Provider-specific names are unresolved until the actual adapter lands; never invent them. Settings: `ap_tolerance_pct`, `ap_tolerance_abs_cents`, `ap_auto_approve_max_cents`, `ap_payment_days_before_due`, `collections_after_days_overdue`, `collections_fee_cents`, `collections_agency_after_days`, `fx_revalue_month_end`, `ledger_accounts`. A payment/export file is not bank acceptance or a filed tax return.

Shipper API keys are scoped per customer and revoked in Portal → API. Keep `WEBHOOKS_ALLOW_PRIVATE` unset/false in production; it exists only for isolated tests. EDI: `EDI_SFTP_ENABLED`, `EDI_POLL_SECONDS`, `EDI_OUR_UNB_ID`; per-partner `EDI_SFTP_PASSWORD_<PARTNER>`. AS2: `EDI_AS2_ENABLED`, `EDI_AS2_ID`, `EDI_AS2_KEY_PEM`, `EDI_AS2_CERT_PEM`, `EDI_AS2_KEY_PASS`; configure trusted partner certs and prove signed/encrypted MDN exchange. HTTP gateway signatures and replay protection remain enabled.

Recovery, backups and release checks

For incidents, capture redacted job/transaction identifiers and UTC sequence, halt relevant automation, determine the first failed boundary, and reconcile external effects before retry. Keep the previous deployment revision and a restorable DB backup. A source rollback does not roll back data or provider actions. After recovery, prove one complete controlled business journey and one relevant negative case; then restore authorised switches deliberately.

Backup frequency, retention, encryption, restore point and recovery time are **not yet verified with Emergent**. Assign an owner to obtain those controls and run a restore test before commercial launch. Do not claim a backup exists solely because hosting is paid.

Source coverage and remaining merge work

Read sources: all sections of `shared/interfaces.md`, `shared/capmap/CAPABILITY-MAP.md`, `shared/s14/README.md`, and s11 go-live documents for Tulli, NCTS/ICS2, AEO, TARIC, origin and controls. The task-tree README discovery also found release-evidence READMEs (`s1-ship-4013082`, `r4-ship-04af356`, `r5-ship-2ce7003`, `r6-ship-a86ad24`) and an EORI capture README; these establish historical evidence, not current activation. Other steps had no step README at discovery. Their declared variables are above and remain subject to merged-code reconciliation. `CONFIG-INVENTORY.md` is complete for the specified baseline, not future revisions.

The old capability map is a code audit, not a live acceptance record. s4 changes legal publication/documentation only; it does not upgrade customs, insurance, messaging or finance capability scores. Complete final-run evidence before upgrading those rows.

Owner activation actions

25 October 2026. This is the company-action checklist for Huolitsija. It does not certify provider activation. Last s4 production identity probe: r8a `fc211b4ec3514c733c719367881563c2f5858004`; provider activation and recorded journey are separate evidence gates. Reuse existing accounts and stored credentials. Only the integrator changes Emergent configuration.

Postmark: account approval and sender authentication

Owner action: in the existing Postmark account, the Account Owner submits the approval request describing transactional freight quotes, booking updates and invoices, expected volume and recipient relationship. If already submitted, supply the information requested by the reviewer; do not create another account. In Sender Signatures/Domains, open heliosone.fi and have its DNS administrator publish the exact DKIM TXT and custom Return-Path CNAME generated for this account. Recheck both verification indicators. DNS authentication and account approval are separate gates. Approval procedure, domain verification API.

Existing DNS action already on the board: NS1 zone `heliosone.fi`, issue `issue_iicklrdl5uct557xuvaeug5jne`. Add TXT at `20261024191646pm._domainkey.heliosone.fi` using the exact public DKIM value in that issue/Postmark DNS Settings; add CNAME `pm-bounces.heliosone.fi` → `pm.mtasv.net`. Leave existing SPF and DMARC unchanged. The domain is already added in Postmark. s14 will verify and inspect a real receipt for `dkim=pass` and `dmarc=pass`; no duplicate request is needed.

Integrator action: retain the existing server token, select `EMAIL_PROVIDER=postmark`, use the verified sender, and configure authenticated inbound and delivery/bounce webhooks. For the email-channel lane, use its final merged endpoint/authentication contract and `EMAIL_CHANNEL_ADDRESS`; do not reuse an obsolete HMAC-only instruction or put a token into public evidence. Prove unauthenticated inbound rejection before enabling forwarding.

Acceptance: an allowed external mailbox receives the actual quote/invoice, its Message-ID correlates to the app, and the provider delivery event is recorded. Approval is needed for recipients outside verified domains. A controlled heliosone.fi-only journey can prove that limited scope while approval remains pending. A blackhole delivery or log row does not prove receipt. No new owner credential request is needed merely to read a verification email; coordinator relays sender/subject issues.

Stripe: live business activation

Owner action: open the existing Stripe Dashboard business onboarding, enter the incorporated legal entity, Business ID, representative/beneficial-owner details requested by Stripe, company bank account and customer-facing support/statement details. Complete requested identity checks and accept the service agreement; resolve any outstanding requirements. Stripe business verification.

Integrator action: after service activation, provision the live key securely and register the live `/api/webhooks/stripe` endpoint with its matching signing secret. Check app `STRIPE_API_KEY` and webhook environment agree. Preserve test evidence separately.

Acceptance: an authorised real transaction has matching amount/currency, authenticated payment event, paid invoice, allocation and zero remaining balance. Until company activation, the specifically authorised test payment is a technical proof only; classify live-ready only if the implementation and round trip work and company activation is the sole remaining blocker.

Tulli: customer status, form 934 and certificate

Owner action: obtain the company's EORI and representation mandates. In Tulli's Authorisations and Decisions Service apply for UTU message-exchange customer status for the relevant procedures and roles (message declarant and, if applicable, service provider/software supplier). Identify the sender and software. Arrange procedure-specific consultation/testing and any required deferment authorisation; the current introduction specifically names payment deferment for import message customer status.

Form 934 distinction: the older official Tulli notice directs certain other-system applications to customs form 934. It is a message-customer application, **not a certificate**. The supplied introduction dated 17 April 2026 directs UTU registration online and other-system applications through the current form/instructions (it specifically mentions Intrastat). Use the current procedure-specific application or obtain Tulli confirmation before submitting 934; do not apply an old ELEX/NCTS list indiscriminately to migrated UTU procedures. Historical official notice.

Certificate action: the authorised company representative applies at DVV e-services for the certificate appropriate to Tulli message signing and client authentication. Have the technical custodian install the issued certificate and private key securely, record expiry and renewal responsibility, and complete signed connectivity testing. The supplied 25 June 2026 technical guide §10.1–10.2 specifies DVV application/installation and a one-year client certificate; notification/GSI server-certificate validity differs. Never publish the key, certificate binding or raw licensing/credential material.

Integrator/customs action: install `TULLI_CLIENT_CERT`, `TULLI_CLIENT_KEY`, `TULLI_DECLARANT_ID`, correct endpoint and procedure modes only after the actual transport passes. Run the agreed customer test cases and retain Tulli's registration decision and authorised production start date. Source: supplied `tulli-introduction.pdf`, printed pp.4–6; `tulli-technical-guidebook-2026-06-25.pdf`, §§10–12. The transport documents are already supplied; do not request them again.

Implementation evidence: the customs lane reports the existing v1.21 transport in `9258593` passed 40 protocol plus 5 exchange-durability checks; `9d7e95f` changes tests/evidence only. Company transport configuration and live authority acceptance remain unverified.

Acceptance: correlated authority acceptance/release for a legitimately authorised declaration. EORI, a local XSD pass or a certificate alone does not prove transport/customer conformance. Missing implementation remains partial; only a working integration blocked solely on registration/certification is live-ready. Customer POA and the owner's standing filing delegation remain separate from technical registration.

Company bank agreement

Owner action: with the chosen company bank, sign a corporate payment/file-transfer agreement covering authorised debit accounts, payment limits, pain.001 submission and pain.002/camt.054 reports. For the implemented Nordea Web Services route, contact Nordea Business Centre, name the administrator, receive the agreement customer/user identifiers and securely enrol the company's PKI certificate using the bank's activation procedure. Arrange renewal and signing authority. Nordea Web Services.

Integrator/finance action: provision the agreed identity/certificate and bank endpoint, verify the bank response-signature trust configuration, and select the implemented API mode only after that check. A shared bank demo certificate must never be used as the company's production identity. Finance confirmed these production settings from `7ade205`: `BANK_PAYMENT_MODE=api`, `BANK_API_PROVIDER=nordea`, `BANK_API_CUSTOMER_ID`, `BANK_API_TARGET_ID`, and `BANK_API_ENVIRONMENT=PRODUCTION`. Supply `BANK_API_CERT` + `BANK_API_KEY` (`BANK_API_KEY_PASSWORD` if encrypted), or `BANK_API_P12` + `BANK_API_P12_PASSWORD`. Optional settings: `BANK_API_URL`, `BANK_API_SOFTWARE_ID`, `BANK_API_FILETYPE_PAIN001`, `BANK_API_FILETYPE_PAIN002`, `BANK_API_FILETYPE_CAMT054`, and `BANK_API_RESPONSE_CERT` (the public Nordea pin is bundled by default). `BANK_API_DEMO` must be false/unset. Keep `HUOLITSIJA_ALLOW_BANK_UPLOAD` unset and never run the bank-upload test.

Implementation evidence: finance reports response-signature verification closed in `23b358c`: bank certificate pin, signed timestamp, and request/customer/environment binding are checked before decoded content is used. An unverified upload response holds the run and blocks resend/cancel pending reconciliation. References: `evidence/s13-continuation/bank-signature-report.md` and `VAT-CONSUMER-HANDOFF.md`. Deployment and company payment remain unverified; this is not a new missing-code claim.

Acceptance: an authorised company payment is accepted by the bank and its genuine status/debit report reconciles once to the correct payable. A generated payment file or a downloaded exchange-rate file is not payment proof. Code/signature gaps remain partial even after the agreement is signed.

Terminal49 paid access

Owner action: open the existing Terminal49 account and agree a paid plan with API read access and the webhook features the business needs; confirm quota, billing and permission to track the company's cargo with the account manager. Reuse the stored key; do not buy a second subscription. Trial and paid transition.

Integrator/visibility action: configure the existing key and authenticated webhook, then use one genuine bill of lading/container reference with customer authority. Previous real reads returned a paid-plan restriction; request-creation guards do not prove tracking.

Acceptance: actual provider read and correlated event appear on that shipment. First real request also requires a real shipment reference. A trial description does not establish this account's entitlement.

AISStream access

Owner action: finish the existing GitHub-authenticated AISStream account access/2FA and create a server-side API key at AISStream Account. Store it through the project's secure credential mechanism. This is an access step, not necessarily a paid contract; do not invent a purchase requirement.

Integrator/visibility action: set `AISSTREAM_API_KEY` on the backend, subscribe to an authorised area/vessel set and confirm actual stream messages. Keep keys out of browser connections. Official stream documentation.

Acceptance: timestamped real vessel observations render on the deployed map with provenance and freshness. A vessel observation does not prove a particular shipment is aboard.

DHL key activation

Owner action: in the existing DHL Developer Portal app, confirm access to **Shipment Tracking – Unified**, complete any account/application activation and ask DHL API support to resolve the stored key's 401 if it persists. Specify that product; Freight booking credentials are separate. DHL Unified Tracking.

Integrator/visibility action: use the stored `DHL_API_KEY`; make one bounded read within the assigned limits and record the status without headers/key. Then use a genuine authorised tracking reference. The prior lane reported 401; recheck the latest lane evidence before treating it as current.

Acceptance: real tracking data enter the correct shipment and persist. A portal key being visible is not API activation.

Carrier DCSA onboarding

Owner action: choose each actual contracted ocean carrier and request its production Track & Trace API onboarding/data entitlement, supported DCSA version, customer identifiers, authentication method, endpoints, quota and webhook arrangements. Separately agree booking authority and commercial rates if bookings are required. DCSA Track & Trace is a standard, not a universal production carrier account.

Integrator/execution action: configure the matching carrier adapter and map real references; verify one carrier event and one booking confirmation where contracted. Keep DCSA reference-server fixtures labelled as protocol checks.

Acceptance: a real carrier acknowledges the authorised booking and sends associated tracking events. A conformant synthetic event does not prove onboarding, capacity or movement.

Redkik contract

Owner action: contact Redkik for the Finnish company's intended countries, cargo/modes, volumes and embedded-insurance role. Agree the actual insurer/intermediary arrangement, territorial/product limits, premium collection, delegated binding/claims authority, data terms and production API onboarding. Sign the resulting company contract; do not infer underwriting authority from a developer key.

Integrator/commercial action: configure the contracted provider credentials/endpoints and delegated limits. Check a genuine quote, authorised bind, policy/certificate and provider reference against the contracted product. Current evidence shows only adapter/refusal/test-double checks, so the capability remains partial until real integration behavior is proven.

Acceptance: insurer-backed offer/binding and delivered policy evidence. Owner contract alone cannot validate the adapter or confer authority beyond the contract.

Creditsafe contract

Owner action: agree a Creditsafe Connect production licence for the required countries/reports, decision use, retention and sharing rights. Ask the account manager to activate the existing account and production entitlement; complete the Welcome Email activation. If that email is in marc@heliosone.fi, relay its sender/subject to the coordinator. Connect onboarding.

Integrator/commercial action: configure the selected credit adapter using the licensed production account. Confirm a real company report and timestamp feed the actual credit decision; preserve absent/unsupported fields as missing.

Acceptance: sourced bureau response with licence-permitted use and observed credit decision. No contract or real API-response evidence exists in the recovered baseline; keep partial, not automatically live-ready.

Completion record

For every item retain date, responsible role, provider/account identifier without secrets, scope, contract/approval reference, configured release, genuine receipt and observed business result. Record company action and software gap separately. Existing sign-ins and authorisations are reused. Do not claim incorporation, a signed DPA, bank activation or provider production access from this checklist being filled in.

Configuration inventory — source snapshot

Extracted from runtime code at base d78217d plus s4. Environment names only: no secret values. See RUNBOOK.md for live transitions and branch-only variables. Source references locate the owning behavior.

CRM

Environment variable	Owning code
CRM_EMAIL_FROM	backend/services/prospecting.py:311
CRM_EMAIL_PROVIDER	backend/services/prospecting.py:307
CRM_IMAP_FOLDER	backend/services/prospecting.py:670;backend/services/prospecting.py:689
CRM_IMAP_HOST	backend/services/prospecting.py:616;backend/services/prospecting.py:655
CRM_IMAP_PASSWORD	backend/services/prospecting.py:663
CRM_IMAP_PORT	backend/services/prospecting.py:656
CRM_IMAP_SECURITY	backend/services/prospecting.py:655
CRM_IMAP_USER	backend/services/prospecting.py:616;backend/services/prospecting.py:663
CRM_OPTOUT_SECRET	backend/services/crm.py:378
CRM_PROSPECT_DAILY_AT	backend/services/prospecting.py:926;backend/services/prospecting.py:938;backend/services/prospecting.py:960
CRM_PUBLIC_BASE_URL	backend/services/crm.py:398
CRM_REPLY_TO	backend/services/prospecting.py:354;backend/services/prospecting.py:355;backend/services/prospecting.py:372
CRM_SANDBOX_DOMAIN	backend/services/prospecting.py:299;backend/services/prospecting.py:921;backend/services/prospecting.py:922
CRM_SMTP_HOST	backend/services/prospecting.py:315

Carriers and tracking

Environment variable	Owning code
DCSA_<SCAC>_API_KEY	backend/services/tracking.py:96;backend/services/tracking.py:112

Environment variable	Owning code
DCSA_<SCAC>_API_SECRET	backend/services/tracking.py:113;backend/services/tracking.py:114
DCSA_<SCAC>_BASE_URL	backend/services/tracking.py:93
CARRIER_BOOKING_MODE	backend/services/carrier_booking.py:101
CARRIER_CONFIRM_HOURS	backend/services/agents/carrier_agent.py:16
DCSA_<SCAC>_KEY_HEADER	backend/services/tracking.py:112

Company identity

Environment variable	Owning code
COMPANY_ADDRESS	backend/routes/legal.py:23;backend/core/config.py:44
COMPANY_BIC	backend/core/config.py:48
COMPANY_BUSINESS_ID	backend/core/config.py:41
COMPANY_EMAIL	backend/core/config.py:45
COMPANY_EORI	backend/core/config.py:43
COMPANY_IBAN	backend/core/config.py:47
COMPANY_NAME	backend/core/config.py:40
COMPANY_PHONE	backend/core/config.py:46
COMPANY_PRIVACY_EMAIL	backend/routes/legal.py:25
COMPANY_REGISTERED	backend/routes/legal.py:15
COMPANY_REGISTERED_OFFICE	backend/routes/legal.py:27
COMPANY_VAT_ID	backend/core/config.py:42

Compliance and privacy

Environment variable	Owning code
PRH_BO_API_URL	backend/services/beneficial_owners.py:111
PRH_BO_CLIENT_ID	backend/services/beneficial_owners.py:112
PRH_BO_SERVICE	backend/services/beneficial_owners.py:114
PRH_BO_USER_ID	backend/services/beneficial_owners.py:113
PRIVACY_RATE_LIMIT_PER_EMAIL_HOUR	backend/routes/privacy.py:60

Environment variable	Owning code
PRIVACY_RATE_LIMIT_PER_IP_HOUR	backend/routes/privacy.py:61

Credit

Environment variable	Owning code
ASIAKASTIETO_BASE_URL	backend/adapters/credit_bureau.py:464
ASIAKASTIETO_PASSWORD	backend/adapters/credit_bureau.py:466
ASIAKASTIETO_USER_ID	backend/adapters/credit_bureau.py:465
CREDITSAFE_BASE_URL	backend/adapters/credit_bureau.py:409
CREDITSAFE_PASSWORD	backend/adapters/credit_bureau.py:411
CREDITSAFE_USERNAME	backend/adapters/credit_bureau.py:410
CREDIT_BUREAU	backend/adapters/credit_bureau.py:392

Customs

Environment variable	Owning code
TULLI_CLIENT_CERT	backend/core/config.py:91
TULLI_CLIENT_KEY	backend/core/config.py:92
TULLI_DECLARANT_ID	backend/core/config.py:93
TULLI_ENDPOINT	backend/core/config.py:90
TULLI_MODE	backend/core/config.py:89
TULLI_SANDBOX_DECISION_SECONDS	backend/services/tulli_sandbox.py:22
TULLI_SANDBOX_RELEASE_SECONDS	backend/services/tulli_sandbox.py:21

EDI

Environment variable	Owning code
EDI_AS2_CERT_PEM	backend/core/config.py:116
EDI_AS2_ENABLED	backend/core/config.py:113
EDI_AS2_ID	backend/core/config.py:114
EDI_AS2_KEY_PASS	backend/core/config.py:117
EDI_AS2_KEY_PEM	backend/core/config.py:115

Environment variable	Owning code
EDI_OUR_UNB_ID	backend/core/config.py:112
EDI_POLL_SECONDS	backend/core/config.py:111
EDI_SFTP_ENABLED	backend/core/config.py:110

Email

Environment variable	Owning code
EMAIL_FROM	backend/core/config.py:72
EMAIL_PROVIDER	backend/core/config.py:71
POSTMARK_SERVER_TOKEN	backend/core/config.py:73
POSTMARK_WEBHOOK_TOKEN	backend/core/config.py:78
RESEND_API_KEY	backend/core/config.py:74
RESEND_WEBHOOK_SECRET	backend/core/config.py:76

Insurance

Environment variable	Owning code
INSURANCE_PROVIDER	backend/adapters/insurer.py:372
REDKIK_BASE_URL	backend/adapters/insurer.py:82
REDKIK_CLIENT_ID	backend/adapters/insurer.py:83
REDKIK_CLIENT_SECRET	backend/adapters/insurer.py:84
REDKIK_COMMODITY_ID	backend/adapters/insurer.py:86
REDKIK_COVER_TYPES	backend/adapters/insurer.py:88
REDKIK_POLICY_ID	backend/adapters/insurer.py:85

OpenAI and legacy AI configuration

Environment variable	Owning code
ANTHROPIC_API_KEY	backend/core/config.py:52
EMERGENT_LLM_KEY	backend/core/config.py:51
LLM_MAX_CONCURRENCY	backend/core/llm.py:162
LLM_MODEL	backend/core/config.py:59

Environment variable	Owning code
LLM_MODEL_FAST	backend/core/config.py:61
LLM_PROVIDER	backend/core/config.py:58
LLM_REASONING_EFFORT	backend/core/config.py:60
LLM_VISION_MODEL	backend/core/config.py:63
LLM_VISION_PROVIDER	backend/core/config.py:62
OPENAI_API_KEY	backend/core/config.py:56
OPENAI_BASE_URL	backend/core/config.py:57

Partner sourcing

Environment variable	Owning code
PARTNER_EMAIL_FROM	backend/services/prospecting.py:311; backend/services/partner_network.py:272
PARTNER_EMAIL_PROVIDER	backend/services/prospecting.py:307; backend/services/partner_network.py:383; backend/services/partner_network.py:951; backend/routes/partner_network.py:27
PARTNER_IMAP_FOLDER	backend/services/partner_mailbox.py:79; backend/services/partner_mailbox.py:104; backend/services/partner_network.py:956
PARTNER_IMAP_HOST	backend/services/partner_mailbox.py:64; backend/services/partner_network.py:955; backend/services/partner_network.py:956; backend/services/partner_network.py:960
PARTNER_IMAP_PASSWORD	backend/services/partner_mailbox.py:72
PARTNER_IMAP_PORT	backend/services/partner_mailbox.py:65
PARTNER_IMAP_SECURITY	backend/services/partner_mailbox.py:64
PARTNER_IMAP_USER	backend/services/partner_mailbox.py:72; backend/services/partner_network.py:955
PARTNER_INBOUND_REQUIRE_AUTH	backend/services/partner_network.py:505
PARTNER_INBOUND_SECRET	backend/routes/partner_network.py:154
PARTNER_NETWORK_KILL	backend/services/partner_network.py:117
PARTNER_OPTOUT_SECRET	backend/services/partner_network.py:208
PARTNER_REPLY_TO	backend/services/partner_network.py:276

Environment variable	Owning code
PARTNER_SANDBOX_DOMAIN	backend/services/partner_network.py:264;backend/services/partner_network.py:498;backend/services/partner_network.py:950
PARTNER_SMTP_HOST	backend/services/partner_network.py:384

Payments

Environment variable	Owning code
BANK_WEBHOOK_SECRET	backend/routes/payments_ops.py:65
PAYMENT_WEBHOOK_SIGNING_SECRET	backend/core/config.py:68
STRIPE_API_KEY	backend/core/config.py:66
STRIPE_WEBHOOK_SECRET	backend/core/config.py:68

Platform, operations and shared sources

Environment variable	Owning code
API_PROXY_TARGET	frontend/src/setupProxy.js:5
API_RATE_LIMIT_PER_MIN	backend/core/api_keys.py:49
APP_BASE_URL	backend/core/config.py:32
COMPANIES_HOUSE_API_KEY	backend/services/beneficial_owners.py:182
COOKIE_SECURE	backend/core/config.py:37
CORS_ORIGINS	backend/core/config.py:28
DB_NAME	backend/core/config.py:27
DEFAULT_PAYMENT_TERMS_DAYS	backend/core/config.py:106
DELIVERY_POD_WAIT_HOURS	backend/services/agents/delivery_agent.py:13
DEMO_OPS_EMAIL	backend/core/config.py:97
DEMO_OPS_PASSWORD	backend/core/config.py:98
DEMO_RESET_PASSWORDS	backend/services/seed_core.py:51
DHL_API_KEY	backend/services/tracking.py:190;backend/services/tracking.py:208
DIGITRAFFIC_ENABLED	backend/core/config.py:101
EXCEPTION_DETECTION	backend/services/exceptions_agent.py:386
EXC_REBOOK_HOURS	backend/services/agents/exception_agent.py:22

Environment variable	Owning code
GEOCODER_URL	backend/services/geo.py:46
IATA_DGR_DATA_PATH	backend/services/dangerous_goods.py:56
INVOICE_AUTO_ISSUE	backend/services/billing.py:545
INVOICE_CUSTOMS_WAIT_HOURS	backend/services/agents/invoice_agent.py:14
JOBS_CONCURRENCY	backend/core/jobs.py:49
JOBS_POLL_SECONDS	backend/core/jobs.py:48
JOBS_WORKER_ENABLED	backend/services/monitoring.py:48;backend/core/jobs.py:50
JWT_SECRET	backend/core/config.py:35
MONGO_URL	backend/core/config.py:26
OPENSANCTIONS_API_KEY	backend/services/pep.py:373;backend/services/pep.py:385
OSRM_URL	backend/services/geo.py:44
OWNER_ALERT_EMAIL	backend/core/agent_runtime.py:350
PHOTON_URL	backend/services/geo.py:47
POSITION_TICK_SECONDS	backend/core/config.py:102
PUBLIC_QUOTE_RATE_LIMIT_PER_HOUR	backend/routes/public_quote.py:27
REACT_APP_BACKEND_URL	frontend/src/lib/api.js:9;frontend/src/features/map/ShipmentMap.jsx:12
SEA_AVOID_RED_SEA	backend/services/geo.py:45
SEED_DEMO	backend/core/config.py:96
SESSION_DAYS	backend/core/config.py:36
SIMULATOR_ENABLED	backend/services/simulator.py:24;backend/services/simulator.py:31
SIMULATOR_STEP_SECONDS	backend/services/simulator.py:25
TEST_TENANT_SANDBOX	backend/adapters/credit_bureau.py:57;backend/core/sandbox.py:14
TRACKING_WEBHOOK_SECRET	backend/routes/webhooks_feeds.py:27
TRADE_GOV_API_KEY	backend/services/screening.py:634
TRADE_STATS_SOURCE	backend/adapters/trade_stats.py:87
TRUSTED_PROXY_HOPS	backend/routes/public_quote.py:28

Environment variable	Owning code
WEBHOOKS_ALLOW_PRIVATE	backend/services/webhooks_out.py:64

Quotes

Environment variable	Owning code
QUOTE_APPROVAL_THRESHOLD_EUR	backend/core/config.py:105
QUOTE_CUSTOMS_FEE_VAT_CODE	backend/services/pricing.py:104
QUOTE_CUSTOMS_INCLUDED_ITEMS	backend/services/pricing.py:106
QUOTE_DOMESTIC_LEG_VAT_CODE	backend/services/pricing.py:103
QUOTE_ICS2_FEE_EUR	backend/services/pricing.py:105

SMS

Environment variable	Owning code
SMS_PROVIDER	backend/core/config.py:81
TWILIO_ACCOUNT_SID	backend/core/config.py:82
TWILIO_AUTH_TOKEN	backend/core/config.py:83
TWILIO_FROM	backend/core/config.py:84
TWILIO_MESSAGING_SERVICE_SID	backend/core/config.py:85
TWILIO_STATUS_CALLBACK_URL	backend/core/config.py:86

Persisted app settings

Stored in MongoDB `settings` document `key=app`; env defaults do not overwrite stored values. Set via the authenticated Settings API; autonomy switches require their dedicated endpoints.

Setting	Source default
autopilot_enabled	False
partner_network_enabled	False
partner_target_margin_pct	18.0
partner_max_price_cents	500000
partner_sell_price_cents	610000
partner_approved_lanes	[]

Setting	Source default
partner_daily_send_limit	20
partner_max_rounds	3
partner_rfqs_per_lane	3
partner_counter_anchor_pct	5.0
autopilot_quote_max_cents	1000000
autopilot_min_margin_pct	15.0
seller_business_id	config.COMPANY_BUSINESS_ID
seller_vat_id	config.COMPANY_VAT_ID
seller_iban	config.COMPANY_IBAN
seller_bic	config.COMPANY_BIC
quote_approval_threshold_eur	config.QUOTE_APPROVAL_THRESHOLD_EUR
default_margin_pct	18.0
customs_clearance_fee_eur	65.0
customs_extra_item_fee_eur	6.0
documentation_fee_eur	35.0
payment_terms_days	config.DEFAULT_PAYMENT_TERMS_DAYS
late_interest_pct	10.5
quote_validity_days	14
insurance_rate_pct	0.25
insurance_min_premium_eur	35.0
ai_auto_send_routine_messages	True
ai_reply_requires_approval	True
auto_approve_low_risk_kyc	True
vat_rate_standard_pct	25.5
vat_rate_reduced_pct	13.5
vat_rate_reduced_low_pct	10.0
record_retention_years	6
retention_financial_statements_years	10

Setting	Source default
retention_quote_unconverted_months	24
retention_kyc_years_after_relationship	5
retention_token_days	7
retention_privacy_request_years	3
terms_version	'1.1 · NSAB 2015 (PSYM 2015)'
sms_notifications_enabled	False
autopilot_halted	False
guard_spend_per_day_cents	500000
guard_confidence_floor	0.85
guard_claim_limit_cents	100000
guard_default_credit_limit_cents	500000
insurance_claim_limit_cents	0
owner_queue_sla_hours	24
owner_escalation_hours	4
owner_alert_email	''
customs_filing_delegated	False
customs_delegation_scope	'direct'
customs_confirm_timeout_hours	24
dunning_enabled	True
dunning_reminder1_days	7
dunning_reminder2_days	21
dunning_final_days	35
dunning_collection_days	45
dunning_fixed_compensation_cents	4000
dunning_reminder_fee_cents	0
rate_expiry_warning_days	14
crm_daily_send_limit	20
crm_follow_up_days	4

Setting	Source default
crm_max_follow_ups	2
crm_target_industries	['1623', '2229', '2511', '2599', '2829', '3109', '4669', '4673', '4675', '4690']
crm_target_regions	[]
credit_default_limit_cents	500000
credit_auto_limit_max_cents	5000000
credit_hold_overdue_days	30
insurance_provider	''

Dynamic families (not literal constants)

- CARRIER_<CODE>_{BASE_URL,BOOKING_URL,API_KEY,ENABLED,SANDBOX,PORTAL_ACCOUNT,SERVICE_CONTRACT_REF,CONTRACT_REF}. Required subset depends on integration type (services/carriers.py, services/carrier_booking.py).
- SMTP_{HOST,PORT,USER,PASSWORD,SECURITY}, PARTNER_SMTP_{HOST,PORT,USER,PASSWORD,SECURITY}, CRM_SMTP_{HOST,PORT,USER,PASSWORD,SECURITY}; source core/email.py and channel reader.
- EDI_SFTP_PASSWORD_<PARTNER> is read by per-partner configuration; never store it in a partner record.
- Pending-step variables and settings are listed separately in RUNBOOK.md so a declaration in interfaces.md is not mistaken for a deployed feature.

Personal-data recipients — code and deployment evidence

Source snapshot d78217d + s4; live activation baseline from integrator probe on a86ad24, 24 October 2026 21:53 UTC. Runtime configuration and signed contracts must be checked again before commercial launch. No hosting region, processor legal entity or DPA has been inferred from a product name.

Recipient / technology	Actual source	Data / role	Observed or conditional activation
Emergent hosting	Live host; backend/server.py, core/db.py; deployment handoff	App/database/documents/log hosting, processor role to be confirmed by agreement	Live host observed by integrator. Hosting region and onward processors unverified.
MongoDB	core/db.py (`MONGO_URL`)	Database software	MongoDB through Emergent; **not evidence of an Atlas contract**. Do not list MongoDB Inc. as a separate processor without actual hosting contract.
OpenAI	core/llm.py direct Responses API and config.py	Selected documents, prompts, messages, extraction/ operational decisions	OpenAI reported available by live integrations API. Actual current model and vision path must be rechecked after model-fix lane; no remembered model ID used here.
Stripe	services/payments.py, routes/ invoices.py	Payment contacts, amount, invoice reference; controller/ processor roles vary by purpose	Live app uses test mode. Production onboarding, live payments and contractual role assessment remain outstanding.
Postmark	core/email.py, services/ notify_channels.py	Recipient/sender, message body and delivery/bounce data	Supported; live email remains log at baseline. Account test/ approved state is separate from adapter configuration.
Resend	core/email.py, services/ notify_channels.py	Same email data	Supported alternative; no evidence currently selected.
SMTP/IMAP mailbox host	core/email.py, services/ partner_network.py, services/ crm.py	Mail content, headers, attachments; processor depends on actual mailbox host	Name the contracted mailbox host before enabling. Do not assume Gmail, Microsoft or another vendor from protocol support.
Twilio	core/sms.py, routes/ channels.py	Phone, SMS text, status callback	Supported; live SMS log mode/ disabled. Trial/test credentials do not prove delivery.

Recipient / technology	Actual source	Data / role	Observed or conditional activation
Redkik / actual insurer	adapters/insurer.py, services/insurance.py	Cargo/route/value/customer contacts	No active insurer on live. Sandbox adapter is unconfigured. Assess insurer/intermediary/controller role and distribution contract before live bind.
OpenSanctions	services/pep.py	PEP screening query, names	Optional paid API only when key configured; baseline activation unverified.
PRH/YTJ, Vero public data, VIES/EORI	services/kyc.py, services/credit.py	Business identifiers and validation	Register/authority sources; not automatically processors under an Article 28 contract.
PRH beneficial owner service, UK Companies House, GLEIF, Wikidata	services/beneficial_owners.py, services/pep.py	Ownership/PEP public data and query identifiers	Some sources need contract/key; inspect activation. Avoid claiming public parent-company data proves natural-person beneficial ownership.
Creditsafe / Asiakastieto	adapters/credit_bureau.py	Business credit query/report	Optional; no bureau active in live probe.
Carriers, agents, terminals, warehouse, Tulli, tax authorities	services/carrier_booking.py, tracking.py, tulli.py and domain routes	Shipment contacts, documents and statutory declarations	Recipients can be independent controllers. Sharing is limited to necessary services/duties and actual destination.
OpenFreeMap; OpenStreetMap Nominatim; Photon (komoot); OSRM	`frontend/vendor/map-source/src/ShipmentMap.tsx` lines77–78; `services/geo.py` lines44–47	Map tile requests/browser IP; backend address or coordinate lookup	Actual default endpoint names read from implementation. Operator overrides can change backend recipients; verify configured host and terms. These are recipients, not assumed Article28 processors.

Required contractual register

For each enabled processor record its contracting legal entity, processing purpose/data, DPA version/date, subprocessors, locations, access controls, retention/deletion, breach contact and any Chapter V transfer mechanism/assessment. For independent controllers record purpose and legal disclosure basis. Obtain these from the actual contracts/configuration, not a guess based on headquarters. Public privacy wording deliberately distinguishes deployed infrastructure, optional adapters and separate controller recipients.

Human responsibilities — Huolitsija, Finland

Reviewed against primary sources on 24 October 2026. Scope: a Finnish private limited company arranging business freight, with Autopilot performing routine operations. **Zero employees is compatible with this operating model; zero accountable people is not.** An owner/director or contracted qualified provider can fulfil many responsibilities. The company, its board, customers and authorities retain their legal roles.

This matrix distinguishes a requirement for a natural person, a company obligation that can be delegated/automated, and a prudent product control. An app label `legal_point` does not itself prove that legislation mandates a human click for every shipment. No legal opinion, licence grant or incorporation has been obtained by this task.

Point / kohta	What is legally required	Human action and automation boundary	Evidence to retain
Board / hallitus	Directors must be eligible natural persons. A small board needs a deputy under the statutory composition rule. The board remains responsible for administration and oversight.	Appoint actual directors/deputy; document delegation and supervision. Software cannot hold a board seat. No employment contract is inherently required for a board member.	Register extract, appointments, board minutes and powers. Companies Act, ch. 6 §§8–10; PRH responsibilities.
Incorporation / perustaminen	Founders execute the memorandum; required statements and registration must be signed by the designated people.	Founders and directors use their own authorised identity. Automation may prepare documents, but must not impersonate signers. A company in formation is not yet the registered contracting entity.	Memorandum, signed declarations, accepted registration. PRH guided incorporation.
Representation and signatures / edustaminen	Board representation and any granted representation rights must be valid.	Owner/director grants scoped authority. Routine electronic acceptances may be generated within an authorised process; there is no general rule that every freight contract needs a fresh handwritten signature. Acts requiring a specified personal signature stay with that signer.	Mandate, scope, expiry, identity and audit trail. PRH representation rights.
VAT registration / ALV-rekisteröinti	Register where VAT law requires; assess voluntary registration and special transactions.	Authorised representative registers and maintains tax authority mandates. Return preparation/transmission can be delegated or automated through authorised interfaces. This is an accountability/onboarding point, not a legal requirement to manually approve each invoice.	VAT decision, effective date, tax mandates, filed-return receipts. Vero VAT registration; Vero authorisations.

Point / kohta	What is legally required	Human action and automation boundary	Evidence to retain
Annual accounts / tilinpäätös	Board and managing director, if any, sign financial statements; company governance approves/adopts them as applicable.	An accountant may prepare books and software may reconcile entries. Statutory signers still sign; an auditor is required only when applicable thresholds or other rules require one.	Signed statements, meeting decision, filing receipt. Accounting Act, ch. 3 §7; Companies Act ch. 5.
Customs mandate / tulliedustus	Representative must be empowered and state direct/ indirect representation; declaration responsibility remains with relevant parties.	Customer's authorised representative grants the customer mandate. Huolitsija's authorised officer grants internal standing delegation separately. Routine filing can then be automated within both scopes. An internal toggle is not the customer's power of attorney or a Customs authorisation.	Customer POA, representation type, EORI, internal delegation, filing identity and receipts. Union Customs Code arts. 15, 18–19.
Indirect representation / välillinen edustus	Declarant and represented party may be jointly liable for customs debt.	Confirm mandate, exposure and guarantee with an authorised officer. Product requires owner review when standing scope excludes indirect representation; the law does not itself require a new human approval on every correctly authorised declaration.	Agreement, guarantee and applicable delegation. UCC arts. 77(3), 84.
AML applicability and accountable officer / rahanpesulaki	Duties apply to the businesses enumerated in the AML Act. Freight forwarding alone does not establish obliged-entity status. Insurance distribution, regulated financial services or other activities can change the assessment.	Record an activity-specific applicability assessment. If in scope, designate accountable management/compliance arrangements under ch. 9 §1 and approve risk procedures. Do not call routine KYB approval a universal statutory human duty.	Assessment naming actual services and supervisor; board-approved policy. AML Act 444/2017, ch. 1 §2, ch. 2 §3, ch. 9 §1.
PEP / poliittisesti vaikutusvaltainen henkilö	For an obliged entity, senior-management approval is required for starting/continuing the relevant PEP relationship; source-of-wealth/funds and monitoring requirements apply.	Qualified senior management decides. App's `pep_relationship` hold is retained as policy even while AML applicability is unresolved. A PEP match alone is not proof of unlawful conduct.	Identity match, grounds, management decision, ongoing monitoring. AML Act ch. 3 §13.
Suspicion report / epäilyttävä liiketoimi	If obliged, report suspicious transactions promptly and observe non-disclosure.	Accountable authorised reporter handles FIU/goAML and confidentiality. The legal obligation is reporting by the entity, not a general ban on technical automation; this app deliberately does not file reports automatically.	Restricted report/receipt and decision; never customer-visible tipping-off. Correct basis: **AML Act ch. 4 §§1,4**, not the earlier draft's ch. 9 reporting citation. Act.

Point / kohta	What is legally required	Human action and automation boundary	Evidence to retain
Sanctions / pakotteet	Prohibited dealings cannot proceed; applicable freezes and reporting must be honoured.	A competent person resolves uncertain identity matches and licensing exceptions under app policy. Neither model confidence nor owner approval can legalise a prohibited transaction. Human review of every clean match is not a general statutory rule.	Current list/version, identity evidence, decision and any authority permission. EU sanctions regulation 269/2014 art. 2; Finnish MFA sanctions.
Dual-use / kaksikäyttö	Authorisations and catch-all/ end-use duties apply according to goods, destination, end-user and role. Not every dual-use shipment needs an individual licence: general/global licences and conditions must be assessed.	Export-control specialist establishes classification, licence coverage and restrictions. Software may prepare/check records; only the authority grants a licence.	Technical classification, licence and conditions, end-use evidence. MFA guidance; EU 2021/821.
Dangerous goods / vaaralliset aineet	Applicable duties require trained/qualified people and, for covered road/rail activities, a safety adviser, subject to exemptions.	Engage a qualified adviser/ service provider if needed and assign trained persons for the actual modal functions. Product's per-booking DG gate is a control, not a claim that the exact same personal signature is universally required across ADR, IMDG and air transport.	Qualifications, appointment/ exemption analysis, shipper declaration, acceptance record. Traficom safety adviser.
Insurance and claims / vakuutus ja vahingot	Distribution authority and actual insurer terms determine permitted sales, binding and settlement.	Authorised insurance contact establishes distribution arrangement, insurer authority and delegated settlement limit. `claim_above_insurance_limit` is a contractual/policy gate; no universal statutory euro threshold exists. Redkik sandbox certificates confer no cover.	Insurance/distribution contract, policy, bind receipt, insurer settlement consent. Insurance Distribution Act 234/2018; Insurance Contracts Act 543/1994.
GDPR human intervention / tietosuoja	GDPR art. 22 applies to solely automated decisions with legal or similarly significant effects on natural persons, subject to specified exceptions and safeguards.	Provide meaningful human intervention where required; assess sole traders/personal guarantees. Name an accountable controller contact. A DPO is required only if art. 37 criteria apply, and may be contracted. Do not assume every company credit decision falls outside GDPR.	Assessment, review channel, authorised reviewer, request/ response record, DPIA if required. GDPR arts. 22, 35, 37–39.

Finnish operating rule

Autopilotti hoitaa tavanomaiset toimeksiannot annettujen valtuuksien rajoissa. Omistaja/hallitus vastaa yhtiön hallinnosta ja valtuutuksista. Pätevyyttä edellyttävät tehtävät voidaan antaa sopimuspalveluntarjoajalle. Pakotetta, puuttuvaa lupaa tai puuttuvaa asiakasvaltakirjaa ei voi ohittaa omistajan hyväksyntäpainikkeella. PEP- ja rahanpesuvelvoitteiden soveltuminen arvioidaan todellisen toiminnan perusteella. Työntekijättömyys ei poista näitä vastuita.

Mapping to the application

`backend/core/legal_points.py` and `/api/autopilot/legal-points` enumerate app holds. `customs_filing_not_delegated`, `customs_indirect_representation`, `sanctions_potential_match`, `ru_by_trade`, `dual_use_export`, `pep_relationship`, `claim_above_insurance_limit`, `director_signature`, `dangerous_goods`, `aml_suspicion_report` are control identifiers, not ten independently proven legal mandates for a human click. The runtime catalogue still contains older legal wording; this document corrects its classification without weakening enforcement. Updating that catalogue is coordinated with s2, not silently changing decision gates here.

Before commercial launch: assign the named legal representative, tax/customs delegate, privacy reviewer, applicable DG specialist and insurance contact; preserve their authority and availability. No person has been assigned by this document. Verify the law and authority requirements effective on the actual filing date; do not rely on an unverified claim that a newer regulation repealed the cited UCC articles.

Final live-site run — executable script plan

Status: **plan only; journey not executed by s4 part 1**. Target `https://huolitsija-app.emergent.host`. Evidence root `/shardflux/project/tasks/task_biwogtmco6in3t5owytfysf4hm/evidence/s4/final-run/`.

Preconditions and stop rules

Read `/api/version` immediately before and after each journey; record commit, frontend asset hash, timestamp, viewport and actor. If deployment changes during a journey, invalidate mixed-revision acceptance and rerun on one identity. Merge s2/s3/s5/s6 and required document/customs/carrier/finance work first. Inspect live settings, receipts and populated routes. Existing `scripts/e2e_autopilot.py`, `scripts/e2e_smoke.py`, `scripts/e2e_pay.mjs` and `scripts/sourcing_cycle_e2e.py` are helpers to adapt, not substitutes for the actual customer path.

Sign in as customer, owner and partner using the credential-store `sign_in` tool in their separate browser sessions. Do not copy secrets into scripts, arguments, screenshots or recordings. An earlier statement that demo passwords were already in the person's store was wrong. Report observed sign-in success or the concrete failure. Correct review reference: `evidence/s2-review/r2/REVIEW-r2.md`.

Inbox verification rule: a one-time code or verification link from marc@heliosone.fi is not a stored credential request and must not become a direct request to the person. Raise an issue to the coordinator with the exact sender and subject for relay. Do not click a Cloudflare or other human-verification challenge. Only genuinely missing stored secrets use a single-field secure request.

Final acceptance uses the real provider in production configuration. Each capability row uses exactly **live**, **live-ready**, **partial** or **missing**. `live` requires a real-service receipt and observed business outcome. `live-ready` means implementation works and only a named company step (incorporation, signed contract or payment) blocks live activation. Missing code, uncertain transport and unproven behavior remain `partial` or `missing`, never `live-ready`.

Provider test mode is allowed only where live operation requires a specific company-only step; name that prerequisite beside the result. The originally requested Customs sandbox and Stripe test-payment sequence remains a technical rehearsal, not the final acceptance endpoint. Obtain genuine mandates and controlled counterparties before the production sequence; no invented shipment, fake signature or simulated movement can pass as real freight. Do not charge a real card or contact arbitrary carriers simply to manufacture evidence. Production payment requires the corresponding authorised company account and transaction.

Required ordinary-path outcome: after customer submission and valid standing authorisations, the system advances all routine forwarder work without an employee/operator manually pushing status. The customer, carrier/sandbox driver, payment provider and legally required owner decisions are separate actors and must be counted. Do not silently approve an ops card to make an autonomous journey appear successful.

Journey and actor table

#	Trigger / actor	Real surface or event	Required observed result / proof
1	Customer	Public `/quote` → chosen lane/ cargo/date	Numeric price and currency, included/excluded charges, mode, rate provenance and validity. Record quote ID and first usable result time.
2	Customer + KYC agent	Signup/claim quote, company KYC	Completed business identifiers, source checks, result and decision actor. Missing/ sanctions/PEP cases hold correctly; not a generic green badge.
3	Customer	Accept quote / booking form	Confirm exact amount and terms version; shipment created once, link to quote retained. Capture duplicate click/retry behavior.
4	Customer + document agent	Real shipment documents	Upload invoice/packing list and mandated POA; extracted actual values, validation and generated required transport docs. No invented signature/ certificate.
5	Authorised officer once, then customs agent	Standing delegation + customer POA; customs detail	Delegation actor/scope recorded; declaration submitted through the real configured authority channel, and correlated authority response/ release recorded. If incorporation/authorisation alone blocks production, label the test result and exact company prerequisite. Missing transport implementation remains partial.
6	Carrier agent + contracted carrier	Carrier booking	Specific carrier/ref/route/ contract, one request, correlated confirmation. A log-only adapter does not satisfy confirmation.
7	Customer	Actual shipment map, desktop and mobile	Same shipment/legs/locations, position time, ETA and provenance. Stale/missing position shown honestly; screenshot the populated route.

#	Trigger / actor	Real surface or event	Required observed result / proof
8	Actual provider event → exception agent	Signed tracking exception event	Observe a controlled, authentic provider delay/rollover once; exception appears, agent selects allowed action, customer notification actually reaches controlled inbox. Rebook if rule requires; preserve old/new booking IDs.
9	Carrier/provider	Delivery event + POD	Delivered state requires actual POD fields, timestamp and source. Simulator is labelled as such.
10	Invoice agent	Invoice detail + downloadable PDF	Issued invoice with correct registered/test seller identity, customer, taxable lines, totals, due date and reference; links to shipment. Missing identity stays a failure for commercial readiness.
11	Customer + Stripe	Checkout from actual invoice	Pay displayed amount/currency through the authorised production account. If live onboarding is blocked only on company incorporation/ verification, perform the originally requested Stripe test payment and report live-ready blocked on that exact step. Capture no secret/card data.
12	Stripe webhook + reconciliation agent	Invoice/payment/reconciliation surfaces	Signed event recorded once, invoice paid, amount allocated, balance zero and ledger/ reference links. Reload and confirm persisted state; duplicate webhook changes nothing. A redirect alone is failure.
13	Owner observes only	Autopilot decision/job/audit trail	Step-by-step actors show agent routine execution and identify any owner intervention. Count expected legal decisions separately.

Record request IDs and business IDs in `steps.csv`: run_id, revision, step, UTC start/end, actor_type, actor_label, route, quote_id, shipment_id, invoice_id, event_id, expected, observed, pass_fail, screenshot, recording_timecode, intervention_reason. Do not include private bearer tokens or unnecessary customer personal data.

Random city-pair coverage

At the beginning of the run draw at least **five** cross-continent pairs from the supported city catalogue, with a stored random seed and the candidate pool/hash. Select continents first to cover Europe–Asia, Europe–Africa, Asia–North America, South America–Europe and Oceania–Asia. Then choose a supported city in each continent by the seeded draw. Do not replace a failing lane silently; record all attempts and why any input was rejected.

Read the rate-card/lane query for each sampled pair and store a redacted proof of whether an exact seeded rate exists. At least **two** pairs must have no seeded rate. If the first draw does not include those, continue the draw and retain the rejected coverage candidates. Use coherent cargo/date/Incoterm data. Request quotes in the real public route (not direct DB inserts). Capture amount, currency, mode, line totals, validity, provenance, elapsed time and decision state for all five or more lanes.

A blank, dash, zero invented freight line, perpetual spinner or “none available” is a failure of the required priced-quote result. “RFQ pending” is a truthful interim result but does not pass a priced-quote requirement; follow the real partner request/reply through to an actual price. Indicative estimates must remain labelled estimates and cannot be reported as confirmed carrier tariffs.

Partner-network cycle

1. Record both autonomy switches, approved lanes, daily send limits, negotiation bounds, global halt and actual recipient routing. Confirm PARTNER_SANDBOX_DOMAIN is unset for the production cycle.
2. Enable Sourcing with typed `enable`. Choose a lane needing capacity and trigger/observe the scheduled cycle.
3. Observe discovered candidate + public source + eligibility; actual outbound RFQ arrives at a controlled partner mailbox. Stored outbox alone does not pass delivery.
4. Partner actor replies with a dated rate, currency, equipment/mode, route, inclusions, validity and conditions. Observe ingestion, source message, parsed fields and OpenAI trace through the actual provider.
5. Observe negotiated counteroffer where applicable, partner acceptance, accepted rate card, carrier record and provenance. A new customer quote must consume that rate rather than an old seeded tariff.
6. Exercise opt-out and confirm no follow-up reaches that recipient. Turn sourcing off and demonstrate queued sends stop; keep global halt/guardrail cases in a separate negative-run recording.

Email-only journey to paid invoice

Use a controlled real external sender mailbox and approved production delivery, with CRM_SANDBOX_DOMAIN/PARTNER_SANDBOX_DOMAIN removed only for their corresponding authorised cycles. Start with a new natural-language RFQ plus attachment, in Finnish or English; no portal login by the customer. Preserve mailbox message IDs/headers with addresses redacted for public evidence.

Email RFQ → delivered priced reply → customer booking reply → secure KYC/POA link and completed validation → documents submitted by email or secure link → real Customs response + carrier confirmation (provider test only for a named company-only prerequisite) → email tracking/exception/delivery → delivered invoice and Stripe payment link → customer payment (test only for a named company-only prerequisite) →

delivered paid receipt. Secure one-time KYC and Stripe-hosted payment pages are allowed; portal-account navigation or manual ops advancement is not. Test same-message replay and a different sender attempting to hijack the thread. Neither may create a second booking/payment or reveal another customer's records.

Recording and screenshots

Capture continuous browser video of the customer route and the key owner/partner observations. Keep a second machine-readable event timeline; elapsed provider waits may be trimmed in an edited presentation only if time jumps are labelled and the full recording is retained. Use actual site renders, never fabricated composites. Hide secret input and credential-store interactions; start recording after sign-in, noting that boundary.

Save `customer-journey.webm` (or MP4), `partner-cycle.webm`, `email-only.webm`; numbered PNGs for quote, KYC, booking, documents, customs response, carrier confirmation, map, exception, POD, invoice, Stripe confirmation and reconciliation. Include 1440px and 390px customer views, plus language variants relevant to legal pages. Include the final invoice PDF and redacted event/actor table.

Acceptance and report

Each named output needs a concrete value and observed behavior. Document legal/owner holds as holds; do not claim “no humans” merely because the app is online. Summarise employee/operator interventions, customer actions, carrier actions, legal decisions and sandbox-driver actions separately. The ordinary journey passes only if routine forwarder transitions happen autonomously and financial reconciliation is observed.

Report revision, routes, exact monetary values, IDs, delivery receipts, screenshots, video/timecodes and per-step pass/fail. Mark unexecuted evidence layers **unverified**; capability status is partial/missing unless the precise live-ready condition above is proven. A sandbox result is never a final capability status. Do not seal final done from this plan. s4 part 1 can be sealed only after legal pages are live and the runbook/legal matrix are in `shared/s4/`.

r8 recording execution scope (coordinator clarification, 24 October 23:25 UTC)

The next recording starts only after the integrator reports r8 live. Use the actual customer email (s6), automatic quote (s5), customer acceptance, booking creation, the single real partner RFQ (s3) and invoice/payment link (s13). No real cargo is supplied. Label pickup, tracking and delivery with their actual capability status; do not inject physical events or turn an acceptance-test request into a dispatch order. This is a recording of the available real digital path, not a claim that physical freight moved.

The coordinator relays customer mail through its existing Gmail session. The read-only IMAP attempt using the existing app password was rejected; no repeated login or new credential request is authorised. Postmark DNS and account approval remain separate gates described in OWNER-ACTIVATION.md. s6 owns the trigger and supplies exact Message-ID/thread/quote/shipment/invoice references. Wait for s4's recording-start notice before sending.

Ops authentication reuses the existing credential-store browser session for `ops.demo@huolitsija.fi`. The coordinator confirmed the named environment password is stale: do not retry API password login or request another credential. API observation may reuse that browser session cookie only in memory; never print or persist cookies, authentication state or secrets. Authenticate before recording and dismiss credential dialogs before capture.

The observer must not approve cards, run agents, change settings, issue invoices or forge events. Autopilot is enabled by the established pre-run configuration, observed at start and end. Every business transition is attributed to customer, app agent, carrier/provider or owner. A zero-write observer alone is insufficient to prove zero owner actions: reconcile the server audit/job timeline for the same run IDs and interval. Any manual advance fails the zero-owner criterion.

Retain a continuous WebM from the deployed browser, per-step screenshots, before/after `/api/version` and frontend asset identities, timecodes, a redacted actor table and an explicit missing-step list. If a stage is held, record the hold and its reason. Do not present a prepared script, old video or app route load as the requested recorded journey.

Recorder: `scripts/s4-record-observer.cjs` uses Playwright video on the actual deployed app. Set `EXPECTED_SHA` to the observed r8 commit and `EVIDENCE_DIR` to a new final-run directory; set `BROWSER_CDP_URL` to the existing signed-in browser debugging endpoint and reuse its app cookie only in memory (no password login). Run under a 60-minute command timeout in a background terminal with stdin available. It exits by stdin `{"stop":true}` or its own 55-minute limit, closes its browser and writes the WebM plus JSON timeline. Commands such as `{"route":"/ops/autopilot"}` and `{"scroll":600}` only observe. The script rejects a different deployment, disabled/halted Autopilot and login redirects. It blocks browser writes and never labels the journey passed. Customer/provider effects run through their normal channels in the coordinated s6/s3 lanes. Record the customer mailbox separately through the coordinator's existing session.

Recorder validation in this continuation is syntax-only; execution on r8 is still pending. Do not run it against the old release to imply final-run acceptance.

Coordinator gate: keep Autopilot off during preparation. Only the integrator enables it after exact r8 deployment and the s3 single-RFQ guard are verified, immediately before recording, and notifies s4. Prospecting/sourcing outreach remains log-only. s4 never toggles Autopilot.

25 October master gate: road Step 7 owns legal pages, this runbook PDF and the recording; wait for the master to confirm that the s8 EUR.1 test is finished and Autopilot may be enabled, in addition to the exact revision/ webhook/single-RFQ checks above.

25 October s6 coordination: use existing test tenant `e19d517b-7637-49e9-8403-97fe63db7728`, sender `marc+huolitsija-email@heliosone.fi`, and explicit `test_tenant=true`; do not create another tenant or change Autopilot settings. Authentication stays in the standing browser session, with app cookies used only in memory and never shared through evidence files. No recording or email trigger before the exact final r8, webhook readiness and master clearance gates above.